Cortex v1.0
Comparative evaluation

Seven-category budgeted evaluation - NVIDIA DGX Spark - BF16

Completion report

Run	Completed jobs	Observed wall time
runs-full-budget-v1	39 / 39	7.61 hours

Interpretation

This evaluation uses seeded samples and synthetic fixtures, not all official full test splits. No overall winner, statistical significance, production readiness or training-data independence is established. Only Qwen3-8B-Base and Cortex v1 are compared. Standard completion and deployment behavior have explicitly different prompt formats.

Time-budgeted coverage of all requested categories, not all official full test splits. Seeded up to 50 examples per standard leaf task; 1,280 bilingual synthetic domain cases per model; 120 paired context cases per model plus 40 Cortex-only 64K cases. Standard tasks use matched completion; deployment fixtures use Cortex native chat vs Base plain transcript with identical demonstrations, schema and 128-token cap. Synthetic cases are parameter-held-out, not independently authored. No overall winner. The original runtime budget was reached; the user authorized continuation to finish remaining work without a new hard stop. Completed work remains preserved.

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Models and fixed revisions

mkd-hossain/Keural-Cortex-8B-v1.0

Revision: a592a43fe940299b5a025e5b719459cda97ae1e6

Source: https://huggingface.co/mkd-hossain/Keural-Cortex-8B-v1.0

Qwen/Qwen3-8B-Base

Revision: 49e3418fbbbca6ecbdf9608b4d22e5a407081db4

Source: https://huggingface.co/Qwen/Qwen3-8B-Base

Protocol

BF16 on NVIDIA GB10. Standard task seeds: 0,1234,1234,1234. Seeded random selection uses up to 50 samples per leaf task, frozen before scored inference. The manifest records exact indices and dataset fingerprints; not all official full test splits are evaluated.

All 45 installed KMMLU subjects and 57 MMLU subjects are included. Five shots for KMMLU, HAE-RAE, MMLU and GSM8K; zero for other standard tasks. Sample counts remain visible beside scores.

Standard tasks use matched raw completion. Deployment fixtures use Cortex native chat versus Qwen Base plain transcript. Identical example content, schemas and evidence are supplied, but serialization differs explicitly.

Deployment fixtures: 480 routing, 240 abstention/no-tool, 240 grounding, 160 supplied-history multi-turn and 160 instruction-constraint cases per model. Both models use greedy generation, repetition penalty 1.0 and a 128-token cap. Model-default inheritance is disabled explicitly. Strict and semantic formatting scores are separate.

Tool arguments have deterministic checks. Clarification fields are checked for presence rather than full human-rated semantic quality. Multi-turn fixtures supply history rather than executing autonomous live workflows. No external tool actions occur. Cases are synthetic parameter-held-out fixtures, not independent holdout evidence.

40 context cases per budget at 4K, 16K and 32K for each model; 40 Cortex-only 64K cases. Same source text in paired tests, actual input tokens recorded. The extension is unpaired. These are not RULER or LongBench results.

GSM8K uses a verified explicit 512-token cap with greedy decoding and repetition penalty 1.0. Effective model configuration and generations are saved; Cortex RoPE translation is asserted before inference.

The original hard deadline saved partial coverage at 22 completed jobs. The user authorized continuation to finish remaining work without a new automatic deadline. Runtime-extension records preserve prior state and completed-artifact hashes. Pending or interrupted work is not zero accuracy. Raw outputs, manifests and uncertainty summaries remain available for audit. Training-data independence is unproven.

All seven categories

Cortex v1.0 vs Qwen3-8B-Base. Differences are percentage points; no significance claim.

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
kobest boolq	40.0% n=50	48.0% n=50	-8.0 pp	Accuracy
arc challenge	52.0% n=50	58.0% n=50	-6.0 pp	Normalized accuracy
kobest copa	74.0% n=50	68.0% n=50	+6.0 pp	Accuracy
kobest hellaswag	66.0% n=50	62.0% n=50	+4.0 pp	Normalized accuracy
kobest sentineg	92.0% n=50	86.0% n=50	+6.0 pp	Accuracy
HAE-RAE / general knowledge	38.0% n=50	38.0% n=50	+0.0 pp	Normalized accuracy
HAE-RAE / history	50.0% n=50	54.0% n=50	-4.0 pp	Normalized accuracy
HAE-RAE / loan word	82.0% n=50	80.0% n=50	+2.0 pp	Normalized accuracy
HAE-RAE / rare word	70.0% n=50	72.0% n=50	-2.0 pp	Normalized accuracy
HAE-RAE / standard nomenclature	80.0% n=50	78.0% n=50	+2.0 pp	Normalized accuracy
KMMLU / biology	56.0% n=50	50.0% n=50	+6.0 pp	Accuracy
KMMLU / chemical engineering	70.0% n=50	68.0% n=50	+2.0 pp	Accuracy
KMMLU / chemistry	56.0% n=50	58.0% n=50	-2.0 pp	Accuracy
KMMLU / civil engineering	44.0% n=50	46.0% n=50	-2.0 pp	Accuracy
KMMLU / computer science	82.0% n=50	84.0% n=50	-2.0 pp	Accuracy
KMMLU / ecology	66.0% n=50	62.0% n=50	+4.0 pp	Accuracy
KMMLU / electrical engineering	50.0% n=50	48.0% n=50	+2.0 pp	Accuracy
KMMLU / information technology	90.0% n=50	92.0% n=50	-2.0 pp	Accuracy
KMMLU / materials engineering	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
KMMLU / math	50.0% n=50	42.0% n=50	+8.0 pp	Accuracy
KMMLU / mechanical engineering	62.0% n=50	64.0% n=50	-2.0 pp	Accuracy
KMMLU / agricultural sciences	54.0% n=50	50.0% n=50	+4.0 pp	Accuracy

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
KMMLU / construction	40.0% n=50	42.0% n=50	-2.0 pp	Accuracy
KMMLU / fashion	54.0% n=50	50.0% n=50	+4.0 pp	Accuracy
KMMLU / food processing	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
KMMLU / health	58.0% n=50	64.0% n=50	-6.0 pp	Accuracy
KMMLU / interior architecture and design	70.0% n=50	70.0% n=50	+0.0 pp	Accuracy
KMMLU / marketing	84.0% n=50	86.0% n=50	-2.0 pp	Accuracy
KMMLU / patent	46.0% n=50	54.0% n=50	-8.0 pp	Accuracy
KMMLU / public safety	42.0% n=50	40.0% n=50	+2.0 pp	Accuracy
KMMLU / real estate	56.0% n=50	62.0% n=50	-6.0 pp	Accuracy
KMMLU / refrigerating machinery	66.0% n=50	64.0% n=50	+2.0 pp	Accuracy
KMMLU / aviation engineering and maintenance	52.0% n=50	50.0% n=50	+2.0 pp	Accuracy
KMMLU / electronics engineering	72.0% n=50	70.0% n=50	+2.0 pp	Accuracy
KMMLU / energy management	40.0% n=50	42.0% n=50	-2.0 pp	Accuracy
KMMLU / environmental science	42.0% n=50	44.0% n=50	-2.0 pp	Accuracy
KMMLU / gas technology and engineering	46.0% n=50	48.0% n=50	-2.0 pp	Accuracy
KMMLU / geomatics	50.0% n=50	50.0% n=50	+0.0 pp	Accuracy
KMMLU / industrial engineer	50.0% n=50	52.0% n=50	-2.0 pp	Accuracy
KMMLU / machine design and manufacturing	56.0% n=50	54.0% n=50	+2.0 pp	Accuracy
KMMLU / maritime engineering	46.0% n=50	54.0% n=50	-8.0 pp	Accuracy
KMMLU / nondestructive testing	62.0% n=50	60.0% n=50	+2.0 pp	Accuracy
KMMLU / railway and automotive engineering	42.0% n=50	46.0% n=50	-4.0 pp	Accuracy
KMMLU / telecommunications and wireless technology	60.0% n=50	56.0% n=50	+4.0 pp	Accuracy
KMMLU / accounting	52.0% n=50	52.0% n=50	+0.0 pp	Accuracy
KMMLU / criminal law	50.0% n=50	44.0% n=50	+6.0 pp	Accuracy

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
KMMLU / economics	66.0% n=50	70.0% n=50	-4.0 pp	Accuracy
KMMLU / education	82.0% n=50	86.0% n=50	-4.0 pp	Accuracy
KMMLU / korean history	36.0% n=50	34.0% n=50	+2.0 pp	Accuracy
KMMLU / law	42.0% n=50	40.0% n=50	+2.0 pp	Accuracy
KMMLU / management	60.0% n=50	64.0% n=50	-4.0 pp	Accuracy
KMMLU / political science and sociology	60.0% n=50	56.0% n=50	+4.0 pp	Accuracy
KMMLU / psychology	58.0% n=50	60.0% n=50	-2.0 pp	Accuracy
KMMLU / social welfare	80.0% n=50	88.0% n=50	-8.0 pp	Accuracy
KMMLU / taxation	46.0% n=50	48.0% n=50	-2.0 pp	Accuracy
MMLU / abstract algebra	64.0% n=50	64.0% n=50	+0.0 pp	Accuracy
MMLU / anatomy	76.0% n=50	78.0% n=50	-2.0 pp	Accuracy
MMLU / astronomy	88.0% n=50	90.0% n=50	-2.0 pp	Accuracy
MMLU / college biology	96.0% n=50	92.0% n=50	+4.0 pp	Accuracy
MMLU / college chemistry	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
MMLU / college computer science	70.0% n=50	64.0% n=50	+6.0 pp	Accuracy
MMLU / college mathematics	62.0% n=50	62.0% n=50	+0.0 pp	Accuracy
MMLU / college physics	56.0% n=50	58.0% n=50	-2.0 pp	Accuracy
MMLU / computer security	80.0% n=50	84.0% n=50	-4.0 pp	Accuracy
MMLU / conceptual physics	88.0% n=50	88.0% n=50	+0.0 pp	Accuracy
MMLU / electrical engineering	76.0% n=50	76.0% n=50	+0.0 pp	Accuracy
MMLU / elementary mathematics	84.0% n=50	84.0% n=50	+0.0 pp	Accuracy
MMLU / high school biology	96.0% n=50	94.0% n=50	+2.0 pp	Accuracy
MMLU / high school chemistry	68.0% n=50	64.0% n=50	+4.0 pp	Accuracy
MMLU / high school computer science	92.0% n=50	92.0% n=50	+0.0 pp	Accuracy

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
MMLU / high school mathematics	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
MMLU / high school physics	76.0% n=50	74.0% n=50	+2.0 pp	Accuracy
MMLU / high school statistics	84.0% n=50	82.0% n=50	+2.0 pp	Accuracy
MMLU / machine learning	72.0% n=50	70.0% n=50	+2.0 pp	Accuracy
MMLU / business ethics	76.0% n=50	74.0% n=50	+2.0 pp	Accuracy
MMLU / clinical knowledge	74.0% n=50	74.0% n=50	+0.0 pp	Accuracy
MMLU / college medicine	84.0% n=50	80.0% n=50	+4.0 pp	Accuracy
MMLU / global facts	46.0% n=50	46.0% n=50	+0.0 pp	Accuracy
MMLU / human aging	86.0% n=50	84.0% n=50	+2.0 pp	Accuracy
MMLU / management	92.0% n=50	88.0% n=50	+4.0 pp	Accuracy
MMLU / marketing	96.0% n=50	96.0% n=50	+0.0 pp	Accuracy
MMLU / medical genetics	90.0% n=50	88.0% n=50	+2.0 pp	Accuracy
MMLU / miscellaneous	92.0% n=50	92.0% n=50	+0.0 pp	Accuracy
MMLU / nutrition	90.0% n=50	86.0% n=50	+4.0 pp	Accuracy
MMLU / professional accounting	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
MMLU / professional medicine	80.0% n=50	80.0% n=50	+0.0 pp	Accuracy
MMLU / virology	64.0% n=50	64.0% n=50	+0.0 pp	Accuracy
MMLU / econometrics	62.0% n=50	64.0% n=50	-2.0 pp	Accuracy
MMLU / high school geography	94.0% n=50	90.0% n=50	+4.0 pp	Accuracy
MMLU / high school government and politics	94.0% n=50	94.0% n=50	+0.0 pp	Accuracy
MMLU / high school macroeconomics	88.0% n=50	82.0% n=50	+6.0 pp	Accuracy
MMLU / high school microeconomics	98.0% n=50	98.0% n=50	+0.0 pp	Accuracy
MMLU / high school psychology	84.0% n=50	86.0% n=50	-2.0 pp	Accuracy
MMLU / human sexuality	82.0% n=50	78.0% n=50	+4.0 pp	Accuracy

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
MMLU / professional psychology	86.0% n=50	90.0% n=50	-4.0 pp	Accuracy
MMLU / public relations	64.0% n=50	66.0% n=50	-2.0 pp	Accuracy
MMLU / security studies	70.0% n=50	70.0% n=50	+0.0 pp	Accuracy
MMLU / sociology	84.0% n=50	86.0% n=50	-2.0 pp	Accuracy
MMLU / us foreign policy	94.0% n=50	94.0% n=50	+0.0 pp	Accuracy
MMLU / formal logic	68.0% n=50	66.0% n=50	+2.0 pp	Accuracy
MMLU / high school european history	82.0% n=50	82.0% n=50	+0.0 pp	Accuracy
MMLU / high school us history	92.0% n=50	94.0% n=50	-2.0 pp	Accuracy
MMLU / high school world history	92.0% n=50	94.0% n=50	-2.0 pp	Accuracy
MMLU / international law	78.0% n=50	82.0% n=50	-4.0 pp	Accuracy
MMLU / jurisprudence	86.0% n=50	86.0% n=50	+0.0 pp	Accuracy
MMLU / logical fallacies	86.0% n=50	90.0% n=50	-4.0 pp	Accuracy
MMLU / moral disputes	82.0% n=50	90.0% n=50	-8.0 pp	Accuracy
MMLU / moral scenarios	58.0% n=50	56.0% n=50	+2.0 pp	Accuracy
MMLU / philosophy	84.0% n=50	82.0% n=50	+2.0 pp	Accuracy
MMLU / prehistory	90.0% n=50	90.0% n=50	+0.0 pp	Accuracy
MMLU / professional law	52.0% n=50	56.0% n=50	-4.0 pp	Accuracy
MMLU / world religions	78.0% n=50	80.0% n=50	-2.0 pp	Accuracy
hellaswag	66.0% n=50	66.0% n=50	+0.0 pp	Normalized accuracy
winogrande	74.0% n=50	76.0% n=50	-2.0 pp	Accuracy
gsm8k	86.0% n=50	84.0% n=50	+2.0 pp	Exact match (strict)
Four-tool routing (strict)	96.2% n=480	0.0% n=480	+96.2 pp	Constraint pass rate
Four-tool routing (semantic)	96.2% n=480	0.0% n=480	+96.2 pp	Constraint pass rate
Abstention and no-tool (strict)	51.2% n=240	53.8% n=240	-2.5 pp	Constraint pass rate

Benchmark / subtask	Cortex v1.0	Qwen3-8B-Base	Difference	Metric
Abstention and no-tool (semantic)	51.2% n=240	53.8% n=240	-2.5 pp	Constraint pass rate
Tool-result grounding (strict)	100.0% n=240	100.0% n=240	+0.0 pp	Constraint pass rate
Tool-result grounding (semantic)	100.0% n=240	100.0% n=240	+0.0 pp	Constraint pass rate
Multi-turn fixture state (strict)	100.0% n=160	50.0% n=160	+50.0 pp	Constraint pass rate
Multi-turn fixture state (semantic)	100.0% n=160	50.0% n=160	+50.0 pp	Constraint pass rate
Instruction constraints (strict)	95.0% n=160	99.4% n=160	-4.4 pp	Constraint pass rate
Instruction constraints (semantic)	95.0% n=160	99.4% n=160	-4.4 pp	Constraint pass rate
Context 4K (strict)	52.5% n=40	0.0% n=40	+52.5 pp	Constraint pass rate
Context 4K (semantic)	100.0% n=40	100.0% n=40	+0.0 pp	Constraint pass rate
Context 16K (strict)	22.5% n=40	7.5% n=40	+15.0 pp	Constraint pass rate
Context 16K (semantic)	45.0% n=40	95.0% n=40	-50.0 pp	Constraint pass rate
Context 32K (strict)	52.5% n=40	0.0% n=40	+52.5 pp	Constraint pass rate
Context 32K (semantic)	75.0% n=40	77.5% n=40	-2.5 pp	Constraint pass rate
Context 64K (strict) Cortex-only extension	0.0% n=40	Not in matched test	-	Constraint pass rate
Context 64K (semantic) Cortex-only extension	20.0% n=40	Not in matched test	-	Constraint pass rate

Job inventory

Model / comparison track	Task	State	Sampling / settings
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	kobest_boolq	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	kobest_boolq	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	arc_challenge	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	arc_challenge	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	kobest_copa	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	kobest_copa	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	kobest_hellaswag	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	kobest_hellaswag	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	kobest_sentineg	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	kobest_sentineg	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	haerae	completed	n=250; 5-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	haerae	completed	n=250; 5-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	kmmlu	completed	n=2250; 5-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	kmmlu	completed	n=2250; 5-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	mmlu	completed	n=2850; 5-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	mmlu	completed	n=2850; 5-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	hellaswag	completed	n=50; 0-shot

Model / comparison track	Task	State	Sampling / settings
Qwen3-8B-Base Matched raw completion; frozen stratified sample	hellaswag	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	winogrande	completed	n=50; 0-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	winogrande	completed	n=50; 0-shot
Keural-Cortex-8B-v1.0 Matched raw completion; frozen stratified sample	gsm8k	completed	n=50; 5-shot
Qwen3-8B-Base Matched raw completion; frozen stratified sample	gsm8k	completed	n=50; 5-shot
Keural-Cortex-8B-v1.0 Native chat + fixed examples	Four-tool routing	completed	n=480; 0-shot
Qwen3-8B-Base Plain transcript + same fixed examples	Four-tool routing	completed	n=480; 0-shot
Keural-Cortex-8B-v1.0 Native chat + fixed examples	Abstention and no-tool	completed	n=240; 0-shot
Qwen3-8B-Base Plain transcript + same fixed examples	Abstention and no-tool	completed	n=240; 0-shot
Keural-Cortex-8B-v1.0 Native chat + fixed examples	Tool-result grounding	completed	n=240; 0-shot
Qwen3-8B-Base Plain transcript + same fixed examples	Tool-result grounding	completed	n=240; 0-shot
Keural-Cortex-8B-v1.0 Native chat + fixed examples	Multi-turn fixture state	completed	n=160; 0-shot
Qwen3-8B-Base Plain transcript + same fixed examples	Multi-turn fixture state	completed	n=160; 0-shot
Keural-Cortex-8B-v1.0 Native chat + fixed examples	Instruction constraints	completed	n=160; 0-shot
Qwen3-8B-Base Plain transcript + same fixed examples	Instruction constraints	completed	n=160; 0-shot
Keural-Cortex-8B-v1.0 Paired source text; deployment serialization differs	Context 4K	completed	n=40; context 4096
Qwen3-8B-Base Paired source text; deployment serialization differs	Context 4K	completed	n=40; context 4096
Keural-Cortex-8B-v1.0 Paired source text; deployment serialization differs	Context 16K	completed	n=40; context 16384
Qwen3-8B-Base Paired source text; deployment serialization differs	Context 16K	completed	n=40; context 16384
Keural-Cortex-8B-v1.0 Paired source text; deployment serialization differs	Context 32K	completed	n=40; context 32768

Model / comparison track	Task	State	Sampling / settings
Qwen3-8B-Base Paired source text; deployment serialization differs	Context 32K	completed	n=40; context 32768
Keural-Cortex-8B-v1.0 Cortex-only extension	Context 64K	completed	n=40; context 65536